

AUTOMOTIVE INDUSTRY STANDARD

Construction and Functional Requirement for Life Support Two Wheeled Road Ambulance

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT and HIGHWAYS
GOVERNMENT OF INDIA

May 2026

INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of standards and development of test facilities in parallel when the work on the preparation of the standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standards Committee (AISC) vide order No. RT-11028/11/97-MVL dated September 15, 1997. The standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CMVR- TSC). After approval, the Automotive Research Association of India, (ARAI), Pune, being the Secretariat of the AIS Committee, will publish this standard. For better dissemination of this information ARAI may publish this document on their Web site.

In 2023, the Government of India felt that to provide “right to access to emergency medical care” across the nation in spirit of Article 21 of constitution of India, the agency that has developed & successfully provide standards for special purpose vehicle shall include scope of the same. Accordingly, ‘Life Support – Two Wheeled road ambulance’ standard is India unique and formulated based on the need for having two-wheeler ambulances which can have a significant usage in rural and semi urban areas. The standard is formulated in two parts as AIS-209 (Part 1 and 2). Accordingly, the requirements of ambulance accessory unit to be installed/ fitted or attached with Two-Wheeler, retrofitted as Life Support Two Wheeled road ambulance ensuring road worthiness, patient’s safety, resuscitation & safe evacuation to requisite emergency care is laid down under this standard.

The standard provides for approval of a life support two wheeled road ambulance when offered as an OEM product and also for approval of an ambulance accessory unit alone or when it is coupled / fitted with a specified L2 category vehicle to make it life support two wheeled road ambulance by retrofitter(s) as new product.

The AISC panel and the Automotive Industry Standards Committee (AISC) responsible for preparation of this standard are given in Annexure-2 and Annexure-3 respectively.

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Construction and Functional Requirement for Special Purpose Vehicle: Two Wheeled Road Ambulance

1.0 SCOPE

This standard specifies the construction and functional requirements for Two wheeled road ambulance of category L2 as per IS 14272:2011.

2.0 REFERENCE STANDARDS

- 2.1 IS 14272: 2011 – Automotive Vehicles – Types – Terminology
- 2.2 AIS-017: 2000 – Procedure for Type Approval and Certification of Vehicles for Compliance to Central Motor Vehicles Rules
- 2.3 AIS 065- Statutory Plates and Inscriptions for Motor Vehicles, their Location and Method of attachment Vehicle Identification Numbering System

3.0 DEFINITIONS

For the purpose of this standard, the following definition shall apply:

- 3.1 **“Two wheeled road ambulance”** means a road ambulance of L2 category that is used for attachment of Ambulance Accessory Unit as defined in 3.2 which, inter-alia shall be used for transport and / or emergent care of victims.
- 3.2 **“Ambulance Accessory Unit”** means set of medical equipment(s) developed as standalone unit referred as Patient Convener System (having supine patient positioning capacity) that can be attached to primary 2-Wheeler vehicle as defined in 3.1 with minimum specification as defined in 4.1.

This Ambulance Accessory Unit may be manufactured directly by original manufacturers of primary 2-Wheelers on their own platform as well as by Retrofitters Industries by taking 2-Wheelers in compliance with requirement(s) in clause 4.1 of this standard.

Note 2: Ambulance Accessory unit, when driving on road, shall always be fitted on left side of vehicle with respect to driver's position.

4.0 APPLICATION FOR TYPE APPROVAL

- 4.1 Application for Type Approval for **Ambulance Accessory Unit** shall be submitted to the test agency, notified under CMVR 126 when fitted with a representative 2-wheeler with minimum engine power ≥ 14 kW or 30 minutes traction motor power ≥ 11 kW. This power requirement shall only be used for selection of base two wheeled vehicle.
- 4.2 The two wheeled road ambulance shall comply with homologation

requirements given in standards notified under CMVR, 1989, as amended from time to time, and this Standard unless and otherwise it is exempted by this standard. Wherever, there is difference in the homologation requirements given in other standards notified under CMVR, 1989, and this standard, the requirements of this standard will be applicable. Technical information to be submitted by the Retrofitter / OEM (all referred as Manufacturer of Ambulance Accessory Unit) shall be as per Annexure 1.

5.0 GENERAL REQUIREMENTS

5.1 OEM / Retro-fitment for the purpose of two wheeled road ambulance is permitted in a vehicle such that it is either fitted as ambulance accessory unit on rear seat (chair conformation) or attached it as separate unit (Patient Convener System), provided it shall so alter the vehicle that no particulars contained in the type approval certificate of the vehicle obtained by the original vehicle manufacturer are at variance unless permitted by the central government or competent authority.

5.2 Any alteration or modification in original vehicle by Retro-fitment / body building in terms of fitment or attachment of ambulance accessory unit to original vehicle shall be carried out in accordance with sound engineering practices.

6.0 Specific Requirements for Two Wheeled Road Ambulance

6.1 Road Ambulance related requirements:

6.1.1 The original 2-Wheeler vehicle used for road ambulance shall have minimum engine power ≥ 14 kW or 30 minutes traction motor power ≥ 11 kW.

6.1.2 Braking Requirements:

6.1.2.1 The braking performance of the vehicle shall comply with the relevant requirements for motorcycle with sidecar as specified in IS 14664:2010, as amended from time to time.

6.1.2.2 Two Wheeled Road Ambulance shall be equipped with a parking brake system such that in the applied position retention is effected by a mechanical means, and the braking effect of which is effected by the frictional force developed between friction surfaces.

6.1.3 Rear View Mirrors:

Vehicle shall comply to the Mirror installation requirement in accordance with AIS-002 (Part2), as amended from time to time, or installation of camera on left side of ambulance accessory unit in addition to the RVM already installed on the original vehicle.

6.1.4 Wheel Rim:

Wheel (s) of side car referred as ambulance accessory unit shall comply with the requirement of IS 16192 :2014, as amended from time to time.

6.1.5 Gradeability

Vehicle shall comply to the gradeability requirement in accordance with AIS-003: 1999, as amended from time to time.

6.1.8 The installation and fitment of ambulance accessory unit shall be done in such a way that it shall be ergonomically feasible and doesn't cause any discomfort to the rider and the pillion.

6.1.9 The patient carrier shall be medical grade collapsible stretcher transformable into chair, conforming with AIS-125 requirements. The size of stretcher conformation shall comply with minimum dimensions of 1800 X 480 mm and height is less than 380 mm in low position with load bearing capacity of maximum 125 kg. The height is not applicable if mono-block undercarriage is not available and must be constructed that it is detachable from the unit.

It shall be designed to provide that full weight of the patient and the carried stretcher part will only be lifted / carried by the personnel for the minimum period of time.

It shall be so designed to provide that during loading and unloading the maximum burden on any personnel is half of the total weight of patient and stretcher and for the minimum possible time and in optical ergonomic position so that back bending is minimized.

The stretcher shall have a minimum of two quick-release patient restraints.

Stretcher (transformable into chair) shall be provided with suitable lockable mechanism while carrying patient.

6.1.10 Ambulance Accessory Unit aka Patient Convener System (PCS) shall have minimum ground clearance of 165 mm.

6.1.11 The overall width of original vehicle and ambulance accessory unit together shall not exceed 2 m.

6.1.12 All the projection and surfaces on the ambulance accessory unit shall be closed and with respect to corners it shall be rounded off leaving no sharp edges and as per requirement shall be covered by soft material (such as viscoelastic gel, rubber or plastic). If a cap is used, its outer dimension shall not exceed 50 mm.

6.1.13 Patient Convener System (with lying patient) shall be designed to provide access to patient from both head (front) and foot (rear) side having minimum distance between two accesses units shall be 100 mm. Front access shall allow direct visual contact with driver with adjustable

blind or other means to maintain privacy, when required.

6.1.14 Design of Patient Convener System (ambulance accessory unit with lying capacity) shall be in accordance such that top deck of loading area of patient's carrier i.e. stretcher is less than or equal to 750 mm from ground. The loading angle shall be maintained and measured in accordance with AIS-125 (Part-1).

6.1.15 Ambulance accessory unit shall be designed with patient's carrier unit isolated from medical device unit with partitioning wall.

6.1.17 Ambulance accessory unit shall be connected by freely movable attachment(s) to original vehicle & supported by separate wheel of defined size, load index & speed rating matching requirement(s) of ground clearance, total load & maximum permissible speed respectively

6.1.6 Puncture Resistance Tyres shall be provided to prevent avoidable time delays for safe & swift evacuation of patient's otherwise puncture repair kit is an essential accessory.

6.1.17 Stands

Vehicle shall be fitted with both, side stand and the centre-stand and comply with AIS-146 requirements.

Extendable stand shall be fitted in Ambulance Accessory Unit attached as separate patient convener system to original vehicle in order to make it stable/stationary while stabilizing patient as per requirement of emergency care.

6.1.18 Emission

Base vehicle selected for two wheeled road ambulance shall meet emission requirements for respective category of vehicles as specified in CMVR 115, as amended from time to time. Two wheeled road ambulance need not to go into any further emission test.

6.1.19 Coupling Requirements

6.1.19.1 The coupling devices shall be manufactured and attached in accordance with good engineering practice, and must be safe to operate.

6.1.19.2 The coupling devices shall be so designed and manufactured that in normal use, with proper maintenance and the timely replacement of wearing parts, they will continue to function satisfactorily.

6.1.19.3 Every coupling device shall be accompanied by installation and operating instructions giving sufficient information for a competent person to install it on the vehicle and operate it properly. The instructions must be in the official language or languages of the Member State in which the coupling device will be offered for sale.

6.1.19.4 The materials that may be used are those for which the properties relevant to the application are laid down in a standard or those for which the properties are given in the application documentation.

6.1.19.5 All parts of the coupling devices whose failure could result in separation of the two vehicles must be made of steel. Other materials may be used provided equivalence has been demonstrated by the manufacturer to the satisfaction of the Testing Agency.

6.1.19.6 All couplings shall be designed for positive mechanical engagement, and the closed position must be secured at least once by positive mechanical engagement.

6.1.20 Electrical/Electronic signal/power connection

There shall be coupler and wiring harness for electrical and signal connection of Two-wheeled vehicle of L2 Category to Ambulance Accessory Unit.

6.2 Ambulance Accessory Unit related requirements

6.2.1 Patient Carrying Capacity

Patient carrying capacity of Ambulance Accessory Unit (Patient Convener System) shall not be more than one applicable for either in sitting or lying supine position in addition to driver's & pillion seat as shall be declared as seating position on original two wheeled vehicles, by the original vehicle manufacturer during Type approval.

6.2.2 Fitment of fire extinguisher

Vehicle shall be equipped with at least one fire extinguisher with extinguishant capacity of 1 kg.

6.2.3 Electrical Wiring & Power Source

6.2.3.1 Electrical wiring and power source requirements shall be as per 4.3.1 and 4.3.2 of AIS-125 (Part 1).

6.2.3.2 Electrical Wiring shall conform to IS 2465-1984, as amended from time to time.

6.2.3.3 Minimum of 2 electrical points of 12 V shall be fitted in patient's carrier or inside patient's chamber as applicable. Further each electrical socket provided in the patient compartment should be permanently labelled as regards its voltage and amperage.

Electrical installations shall comply with those clauses of IEC 60364-7-708 which are applicable to ambulances.

Note : The reference to IEC 60364-7-708 does not apply to the original electrical equipment, which is already covered by the type approval of the base vehicle.

6.2.4 **Lights/ Emergency Lights/ Warning/ Visibility**

- 6.2.4.1 Vehicle shall be equipped with single blue colour flasher (commonly known as beacon) suitably fixed, preferably center aligned and visible to road user.

All warning lights have to be mounted rectangular to the horizontal ground. They must provide 100% of their intensity in a vertical angle of ± 4 degrees and 50 percent in a vertical angle of ± 8 degrees. The minimum intensity is for blue and red lights at 100 cd at daylight and 200 cd in the night. The horizontal minimum angle should be at least 45 degrees. All lights must flash between 2Hz and 4Hz.

- 6.2.4.2 Illumination of patient's chamber in Patient Convener System (referred as ambulance accessory unit) shall be minimum of 50 lux.

Light levels shall be measured along the central longitudinal axis of the stretcher at the head, mid-point and foot position with the stretcher in its normal position for transportation in the ambulance.

Note: The colour temperature of the light will change the appearance of skin and organs. Therefore, it is important that the interior lighting is suitable for patient care during transport. Although it may not be necessary in ambulance use to define "daylight" or "natural colour balance" in a more exact way other than the colour temperature.

- 6.2.4.3 One direction indicator lamp, each on front and rear side on outer wall of Patient Convener System (PCS) shall be provided in addition to the direction indicator lamp already installed on the original vehicle. It shall be operated through same switch which is available on base vehicle.

- 6.2.4.4 All the lights/ emergency lights & warning beacons shall draw power from replaceable battery provisioned with ambulance accessory unit and shall not draw any power from battery of original vehicle

6.2.5 **Sirens**

- 6.2.5.1 Orientation: siren loudspeakers shall be mounted towards the front. The main sound direction must be in driving direction.

- 6.2.5.2 Other requirements: Permitted are wail and yelp signals that cycle between 10-18 respectively 150-250 per minute at sound pressure level of 110 dB(A) to 120 dB(A). The sound pressure level of sirens should be tested in accordance with IS 1884. The frequency range must be at least one octave and should be between 500 Hz and 2000 Hz. An additional electronic air horn can be used. Further there should be a public address system that can be worked at all times ergonomically from the driver's seat. The siren switch can only be used if the warning lights are on.

- 6.2.5.3 The power for siren loudspeakers and PA system shall be drawn from replaceable battery provided in ambulance accessory unit and not from

battery of primary vehicle.

6.2.6 Patient's Safety requirements

- 6.2.6.1 Head Immobilization system shall be fitted within unit for prevention against unexpected movements during evacuation.
- 6.2.6.2 Body Immobilization with minimum of two quick release restraints for stabilization of patient.
- 6.2.6.3 Pressure dispersing system shall be fitted on stretcher for effective pressure dispersion led comforting of patient while transit.
- 6.2.6.4 Noise suppression system shall be in place for patient for protection against unavoidable noise during transit.
- 6.2.5.5 Emergency buzzer shall be in place within reach of patient's hand to give alert.
- 6.2.6.6 Suitable mechanism shall be in place for fresh air infusion on requirement basis.
- 6.2.6.7 The height of chamber shall be maintained to 700 ± 50 mm to provide feel of openness.
- 6.2.6.8 Tinted laminated windows shall be provided in front and rear access similar to windshield to prevent aggravation of injuries during unprecedented collisions. Access units of either side shall be opened from both inside and outside of patient's chamber.
- 6.2.6.9 Head protection system shall be fitted on front or head side with suitable impact resistance ability.
- 6.2.6.10 Direct observation window shall be in place for direct observation of patient both by driver and medical first responder sitting on front & rear seat of original vehicle respectively.
- 6.2.7 Medical Equipment(s) shall be fitted in accordance with AIS -209 Part II along with additional requirement(s) as deployed. ⁽¹⁾
- 6.2.8 Minimum of 2.2 L water capacity of portable oxygen cylinder support shall be provided along with hook for infusion mounting. ⁽¹⁾
- 6.2.9 Ergonomic provisions for adequate storage and operational space for fitment of Equipment's for continuous monitoring of vehicle as well as vital parameters of patient with telemedicine support fitment's positioning requirement shall be complied though fitting of such Equipment's in accordance to listed equipment(s) in AIS-209 Part II. The vehicle's retrofitter's manual shall include details of space allocated for the same. ⁽¹⁾
- 6.2.10 Continuous oxygen support and requisite diagnostics are also optional choice(s) for prolonged emergency care at site as per requirement of

end-user. The details of items shall be included in retrofitter's manual for selection at the end of user for fulfilling specific requirements in accordance to site of deployment.⁽¹⁾

⁽¹⁾ These provisions shall be approved by RTO in consultation with medical experts at State level to qualify it as "ambulance".

6.2.11 **Battery and alternator**

Batteries shall be positioned to allow maintenance without removing the battery from its securing device. The construction of the battery and all connections to it shall be such as to prevent any possibility of an inadvertent short circuit.

Note 1: Additional batteries may be required to power the medical devices carried on board and the intended use of the ambulance. In such cases, the manufacturer shall ensure optimal charging of the additional batteries without any impact on the primary vehicle battery. The additional circuit shall not draw current more than that specified by the manufacturer. However, if a Genset is provided, requirements of additional batteries shall not be applicable.

6.2.12 **Fire safety**

All interior materials shall comply with the flammability requirements specified in IS: 15061, as notified under CMVR, 1989 though the standard does not cover ambulance in the scope.

6.2.13 EMC/EMI: Wherever vehicle is added with electronic system(s), or ESA they shall comply with requirements specified in AIS-004 (Part 3), as amended from time to time.

7.0 **REQUIREMENTS FOR CONSPICUITY MARKING**

The color, front, rear and side –requirements applicable to Type B Ambulance were adapted with respect to size to fit in space requirements of 2-Wheeler attached patient convener system

Conspicuity Requirement	Standard
Color of Patient Convener System	Basic color of the complete exterior should be brilliant white (RAL code 9010) excluding front, rear and side bumper (if any) and primary vehicle. The color should be weather resistant and withstand daily cleaning and washing*. Chevron pattern: Red/ Yellow for this standard.
Front	Word "AMBULANCE" on yellow background, not less than 250 mm in width, shall be in mirror image

	(reverse reading) for mirror identification by drivers ahead. Length should cover maximum coverage possible of the surface. The size can be reduced if the space is not available however visibility shall not be lost. It shall be in contrasting color and shall be retro-reflective. A retro-reflective tape shall be provided in front.
Rear	Word “AMBULANCE” on yellow background, must be displayed on rear. It shall be in contrasting color and shall be retro-reflective & essential. Below word AMBULANCE will be “Star of life” symbol as 85% of rear display surface with right side having display of emergency number logo with same size. Both of these items shall take equal half of the rear display surface. These items are optional based on size availability. A retro-reflective tape shall be provided in rear side below this content.
Side(s)	The left or visible side of the vehicle (away from driver side) should be equipped with a two lines red retro-reflective batten burn pattern on the white ground color with minimum of 6 X 6 cm should be in middle of left side surface of patient convener system length and end in top square, with an Ambulance marking on top of these marking. The word “AMBULANCE” shall be in contrasting color to the white background and shall be retro-reflective. The front half of the batten burg pattern should be red / yellow squares.
Ambulance Accessory Unit	The Conspicuity improvement requirements for Ambulance Accessory Unit shall be installed based on space availability including color, front, rear, and side –requirements similar to Type B Ambulance however adapted to fit in space restraints.

ANNEXURE 1		
INFORMATION TO BE SUBMITTED BY ORIGINAL EQUIPMENT MANUFACTURER OR RETROFITTER AT THE TIME OF APPROVAL		
Sr. no	General	Details
1.	Retrofitter details	
1.1	Name and address of the customer	
1.2	Name of variants, if any	
1.3	Plant(s) of Manufacture / Retrofitment	
2.0	Description of Submitted Vehicle	
2.1	Category	
2.2	Model	
2.3	Vehicle manufacturer	
2.4	Engine no	
2.5	Chassis no	
2.6	Engine displacement(s)	
2.7	Max Power (bhp@rpm)	
2.8	CMVR certificate no	
2.9	Fuel type	
2.10	Fuel tank capacity	
2.11	Dimensions (mm)	
2.11.1	Length	
2.11.2	Width	
2.11.3	Height	
2.12	Ground clearance	
2.13	Kerb Weight of SPV	
3.0	Life Support Emergency Care System	
3.1	Ambulance Accessory Unit (Rear Seat Chair Conformation/ Patient Convener System-attached unit)	

3.2	Patient Carrying Capacity & Positioning [Sitting/ Lying]	
4.0	Schematic Representation indicating the arrangement, position & dimensions of ambulance accessory unit with primary vehicle	
5.0	Details of alterations done on original vehicle (separate appendix can be enclosed) to fit/ attach ambulance accessory unit	
6.0	Photograph of the vehicle with ambulance accessory unit	

ANNEXURE 2	
(See Introduction)	
COMPOSITION OF AISC PANEL ON	
CONSTRUCTION AND FUNCTIONAL REQUIREMENT FOR TWO WHEELED ROAD AMBULANCE	
Name	Organization
Dr. Amit Gupta (Convener)	All India Institute of Medical Sciences
Mr. A. A. Badusha	ARAI
Mr. Kamalesh B. Patil	ARAI
Mr. Pratik Nayak	ARAI
Mr. Sujit Chinchwade	ARAI
Mr. S. N. Dhole	CIRT
Mr. Deepak Joshi	ICAT
Mr. Gurkaran Singh	ICAT
Mr. Deepak Joshi	ICAT
Mr. Nagarajan S	GARC
Dr. Kailash Manda	---
Dr. Raman Chawla	---
Mr. K. Kapila	IRF
Mr. Balraj Bhanot	IRF
Mr. Parmeshwar Mane	SIAM (Bajaj Auto Ltd.)
Mr. Piyush Chowdhry	SIAM (Hero Moto. Corp. Ltd.)
Mr. Ishan Gupta	SIAM (Hero Moto. Corp. Ltd.)
Mr. Shiva Arun Teja	SIAM (Hero Moto. Corp. Ltd.)
Mr. Apoorva Tripathi	SIAM (Hero Moto. Corp. Ltd.)
Ms. Garima	SIAM (Hero Moto. Corp. Ltd.)
Mr. S. Gururajan	SIAM (TVS Motor Company Ltd.)
Mr. Sanchit Khare	OLA Electric
* At the time of approval of this Automotive Industry Standard (AIS)	

ANNEXURE 3 (See Introduction) COMMITTEE COMPOSITION * Automotive Industry Standards Committee	
Chairperson	
Dr. Reji Mathai	Director, The Automotive Research Association of India
Members	Representing
Representative from	Ministry of Road Transport and Highways
Representative from	Ministry of Heavy Industries
Representative from	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises
Head-TED	Bureau of Indian Standards
Director	Central Institute of Road Transport
Director	Global Automotive Research Centre
Director	International Centre for Automotive Technology
Director	Indian Institute of Petroleum
Director	National Automotive Test Tracks
Director	Vehicles Research and Development Establishment
Director	Indian Rubber Manufacturers Research Association
Representatives from	Society of Indian Automobile Manufacturers
Representatives from	Tractor and Mechanization Association
Representatives from	Automotive Components Manufacturers Association of India
Representative from	Indian Construction Equipment Manufacturers' Association
Member Secretary	
Shri Vikram Tandon	The Automotive Research Association of India

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